

A conducting ring driven through an internal cut

Hybrid $A-\Phi$ finite-element solver, tree-cotree gauge. The first **multiply connected** case in the suite: the current circulates with no terminal anywhere on the outer boundary, so the ring can only be driven by cutting it.

regression_tests/05_Loop_1A_50Hz · 1 A internal current source ·
50 Hz · 509,598 unknowns · backward error 1.0e-22

0. What this case establishes

Cases 01–04 are all a straight rod. A rod is **simply connected** and the tree-cotree gauge has nothing hard to do on one. A ring is **multiply connected**: the cotree has to span the loop, the current has to be able to go all the way round, and there is no outer surface to attach an electrode to. It can only be driven through a **cut**.

The check no other case can make. Inject 1 A across the cut, then measure the current crossing a plane where *nothing is prescribed*:

	MEASURED
I through the cut , where 1 A is injected	1.000503 A
I through $\theta = \pi/2$, nothing prescribed	1.000176 A

The current circulates the whole way round. A gauge that wrongly killed the loop would show up there and in no other case in the suite.

Everything with a closed form, against it

This geometry is unusually well supplied with exact solutions, and §5 compares against all of them. The summary:

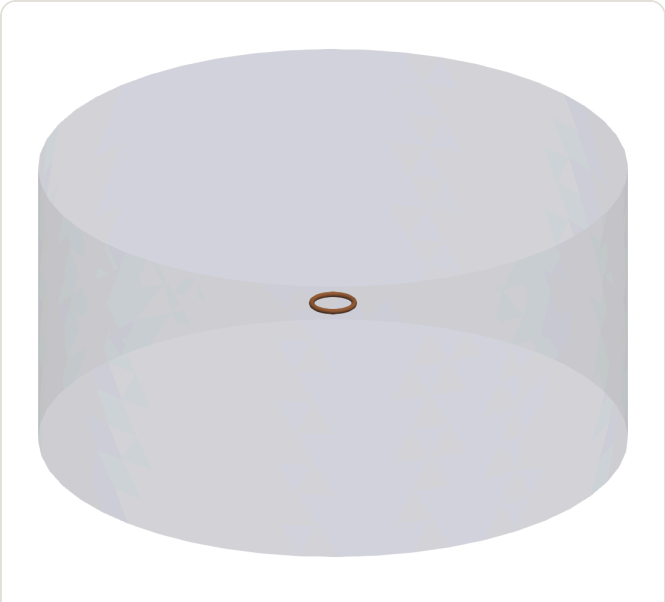
QUANTITY	REFERENCE	ERROR
E inside the conductor	exact, $I / (\sigma \rho K)$	0.016 % median, 0.035 % at the 90th pct
R	exact, $2\pi / (\sigma K)$	-0.128 %
B on the loop axis, at the centre	exact, superposed over the real current	-0.37 %
B in the air, $2a - 20a$	filament, elliptic integrals	inside ± 0.75 %
L	$\mu_0 R [\ln(8R/a) - 2 + \frac{1}{4}]$	-0.09 %

Inside the conductor, where the exact solution is a genuine closed form rather than a thin-wire idealisation, the solver reproduces it to four significant figures in every one of 23,409 elements.

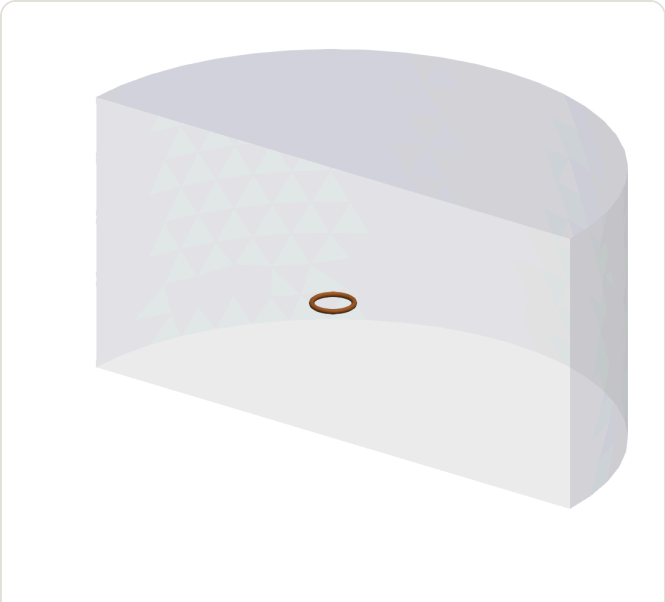
Four of those five used to be 2–25 % out, and it was one cause: the air domain. It stopped 2.3 loop radii from the ring, and `flux_tangential` makes the wall behave like a flux-excluding shell. Everything that touches the field away from the conductor was wrong, all in the direction that picture predicts. The domain is now 90 mm — which costs **fewer** elements, not more, because the grading scales with it — and `loop_small.geo` keeps the 30 mm version as the control.

§5.7 is the argument. §5.6 retracts two earlier tests of mine that said the domain did not matter; both were negative results from experiments that moved more than one variable.

1. The case



What was modelled. The copper ring at mid height inside the air cylinder, drawn transparent. Loop radius 6.5 mm and tube radius 0.8 mm against a domain **90 mm** in radius and 90 mm tall, so the ring is a small object in a large box. That is deliberate and it is §5.7's subject: at the 30 mm this case used to ship, the outer boundary was the largest error in it.



Cut at $y = 0$, the plane the port sits on, so the ring is visible rather than seen through the domain.

Conductor	copper, $\sigma = 5.8e7 \text{ S/m}$; loop radius $R = 6.5 \text{ mm}$, tube circumradius $a = 0.8 \text{ mm}$, a 20-gc
Domain	cylinder of air, 90 mm radius, 90 mm tall, $n \times A = 0$ on the outer boundary
Excitation	1.0 A through the cut at $\theta = 0$, 50 Hz
Skin depth	$\delta = 9.346 \text{ mm}$, so $a/\delta = 0.086$ — no skin effect
Mesh	38,248 nodes, 237,031 elements, Netgen-optimised
Measured	$Z = 3.542415e-04 + 6.215758e-06 \text{ j } \Omega$

1.1 Where the dimensions came from

Not from a document. Two **published** closed forms, solved together against a reference impedance measured for this geometry:

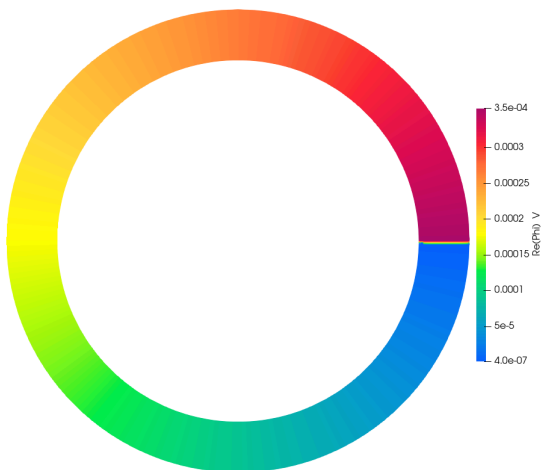
$$R_{dc} = 2 \pi R / (\sigma A)$$

$$L = \mu_0 R \left[\ln(8R/a) - 2 + 1/4 \right] \quad \text{round wire, uniform current}$$

$R = 6.5 \text{ mm}$ with $a = 0.8 \text{ mm}$ reproduces the reference R to 1.3 % and L to 0.27 %, and agrees independently with dimensions measured off the reference model's own scale bar — centreline diameter 1.23 cm, tube 1.4–2.0 mm.

The reference was run at 100 Hz, not 50. That was stated nowhere and falls out of the same two equations. $\text{Im}(Z)/\omega$ is 19.75 nH at 100 Hz and 39.50 nH at 50 Hz, and *no* geometry consistent with the reported R can give 39.50 nH. It does not matter for what this case checks: a/δ is 0.086 at 50 Hz and 0.121 at 100 Hz, so there is no skin effect either way and R and L are frequency independent between them.

2. The potential



Re(Φ) over the whole ring, seen from $+z$. It ramps smoothly all the way round and **falls off a cliff at the cut**, at 3 o'clock.

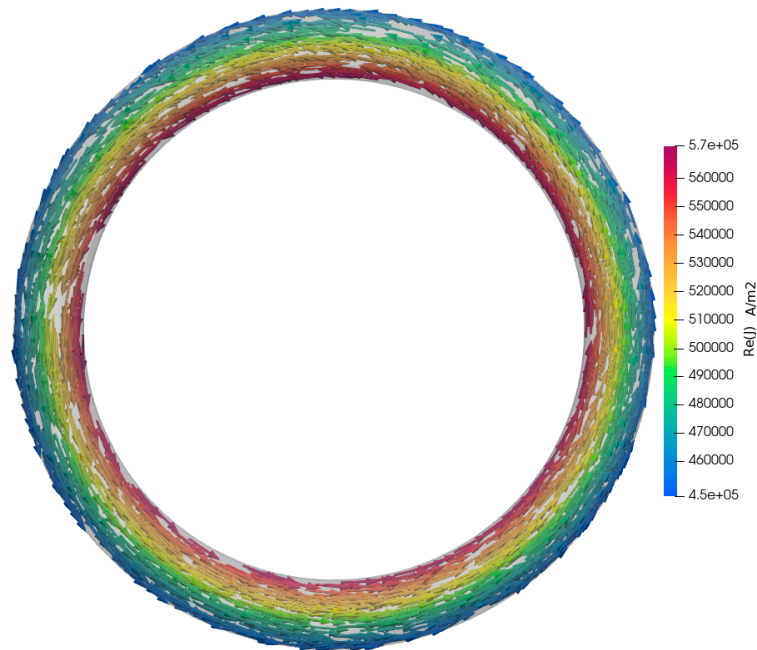


The same, obliquely, so the tube reads as a tube. Flat-shaded: 3D lighting darkens one side by more than Φ changes across a colour band.

This is the picture of what an internal cut actually does, and nothing in cases 01–04 looks like it. One side of the cut is the port's 0 V reference; the other floats to whatever the solve returns. **That jump is the terminal voltage** — $3.542415\text{e-}04 \text{ V}$ here, for 1 A, which is the impedance.

Φ is single-valued on each side: 193 nodes sit exactly on the cut plane and every one reads the same value, so the jump is a genuine discontinuity across a zero-thickness interface, not a gradient through a thin region.

3. The current



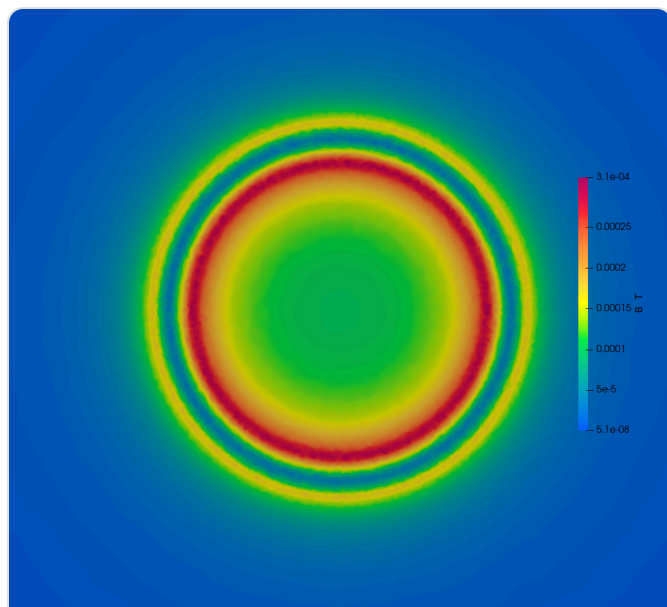
$\mathbf{J} = \sigma \mathbf{E}$ on the ring, every 11th cell, coloured by magnitude. The arrows close the loop.

Look at the colour, not just the arrows. $|\mathbf{J}|$ is highest on the **inner** radius — 5.8×10^5 against 4.4×10^5 A/m², a ratio of **1.32**. That is not noise: current flowing azimuthally takes a shorter path on the inside, so it crowds there. $(R+a)/(R-a) = 1.28$, and the measured 1.32 is that effect.

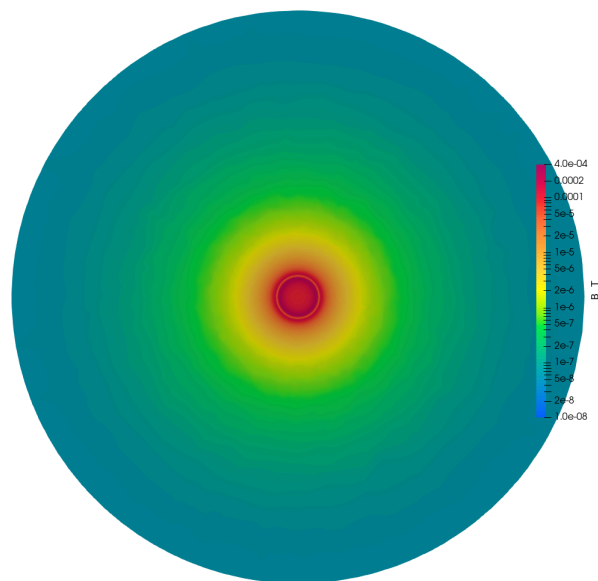
It is also why the exact R below is **not** $2\pi R/(\sigma A)$.

4. The fields

4.1 B in the xy plane



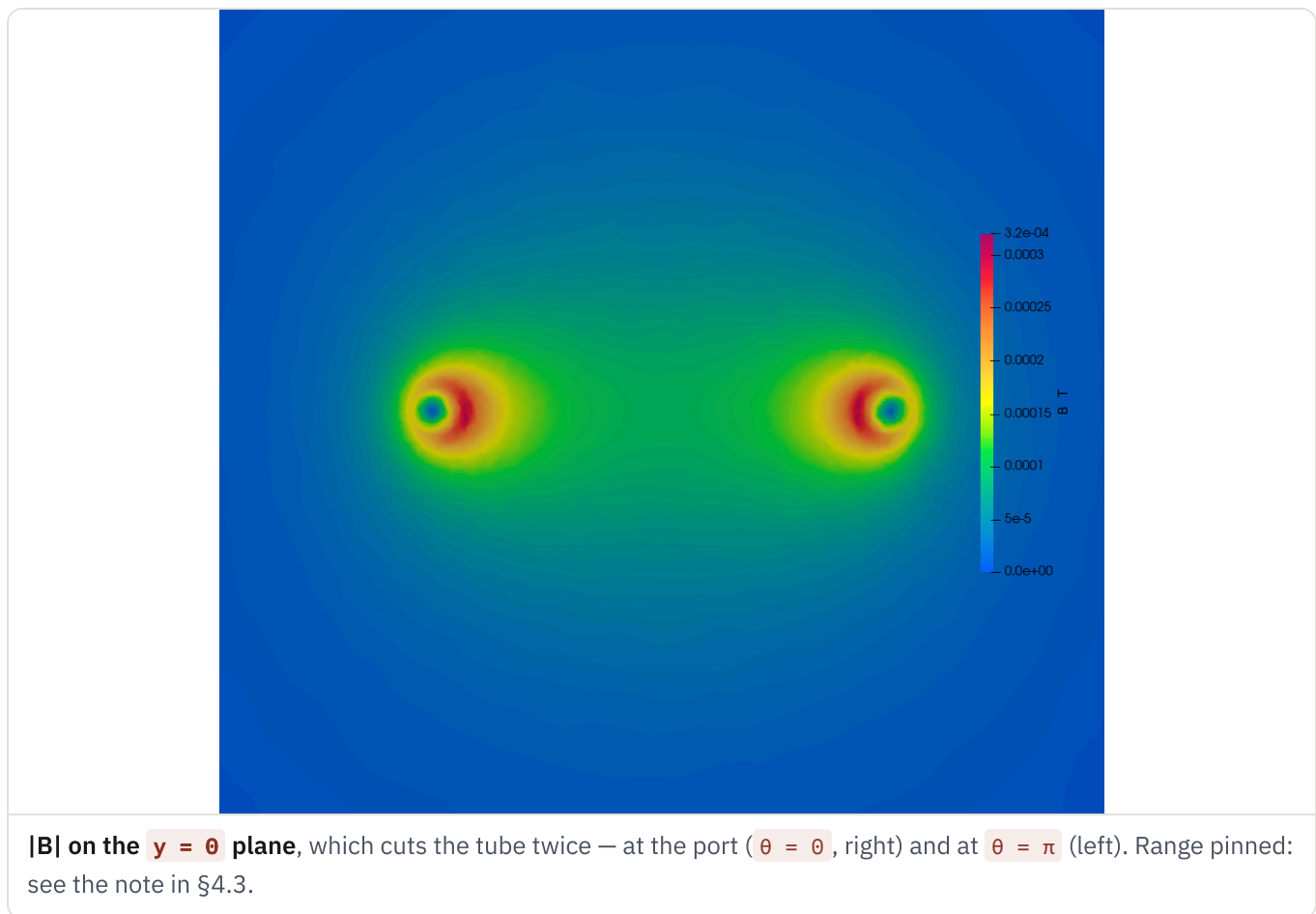
|B| through the tube centreline, zoomed. Red at the **inner** tube surface, a minimum at the tube axis, yellow at the outer surface.



The whole 90 mm domain, log scale and pinned, so the decay to the wall is visible. The field is flat and dark well before the wall is reached, which is the point of §5.7: the boundary is far enough out to stop interfering.

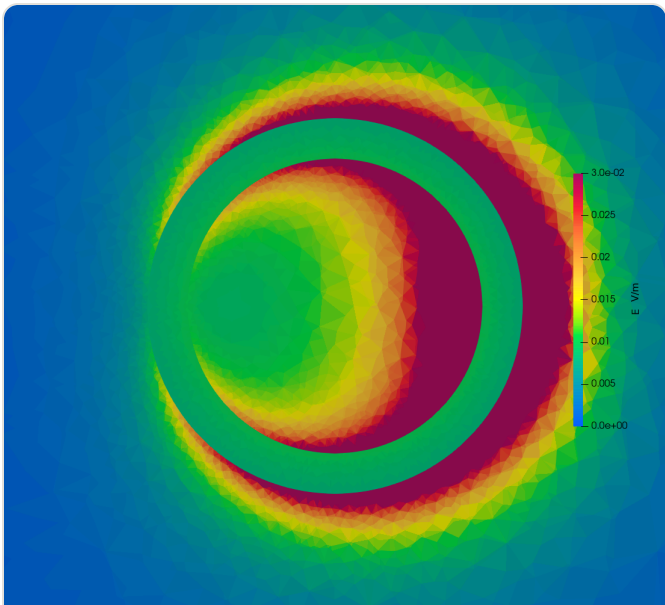
Three things in the zoomed view are worth naming, because each is a check. **|B|** has a **minimum at the tube axis** — the wire's own field vanishes there. It **peaks at the tube surface**, where Ampère gives $\mu_0 I / (2\pi a) = 2.5e-4 \text{ T}$. And the inner surface is **hotter than the outer**, because the rest of the loop's field adds on the inside and subtracts on the outside.

4.2 The port plane

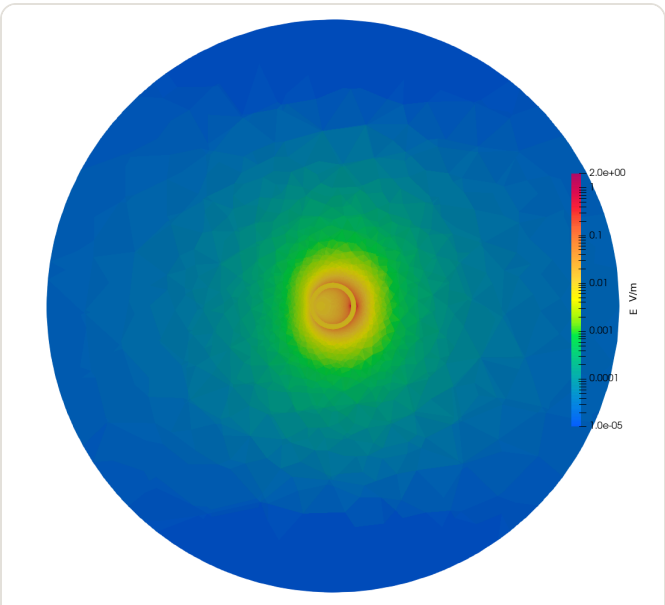


Each conductor carries its own circulating field with a minimum on its axis, and the loop's hole between them is brighter than the outside — the two contributions add in there and oppose outside. This is the plane the cut is made on.

4.3 E in the xy plane, and a warning about colour ranges



|E|, zoomed, range pinned to 3.0e-02. The conductor is the uniform green annulus; the air around it is *higher*.



The whole domain, log scale.

Two earlier claims in this section were wrong and are withdrawn. The first was a table of "median **|E|** inside the conductor / in the air / at the cut" taken as **volume medians over whole regions**. Averaging **|E|** over a region that is not homogeneous does not measure anything: in case 02 the same statistic is dominated by the rod's *end caps*, where **|E|** reaches 197.8 V/m against 25 V/m in the body, and here it mixes the port's neighbourhood with the far side of the ring. The second was the claim that the air's field exceeds the conductor's "in every other case", supported by a cross-suite table built from those same volume medians. **Both are replaced below by measurements made on a stated plane, in a stated shell, on a stated side.**

Measured on the **y = 0** plane — the plane these figures actually cut — split by side:

SHELL AROUND THE TUBE	PORT SIDE, x > 0	JOINT SIDE, x < 0
inside the conductor	0.00867 V/m	0.00864 V/m
air, 1.0–1.5a	0.18829	0.00860
air, 1.5–2.5a	0.08383	0.00685
air, 2.5–5a	0.03321	0.00511

Inside the conductor the two sides agree to 0.3 %, as they must: the same current flows through both. Immediately outside, they differ by **22x**. That difference is the port, and nothing else.

On the joint side the air just outside the tube is indistinguishable from the conductor's own field — 0.00860 against 0.00864, half a per cent apart — and it then decays outward. So there is no general rule that "the air's **E** is larger". There is a boundary condition at the wall, which §5.8 shows this mesh can only just resolve, and a decay away from it.

For comparison, case 02's rod measured the same way, on the mid-height plane its own figure cuts:

	MEDIAN E
inside the conductor	24.935 V/m
air, 1.0–2.5a	24.600
air, 2.5–5a	14.927
air, 5–10a	9.257
maximum anywhere on that plane	25.307

On the plane the figure shows, case 02's air is **not** higher than its conductor — 24.6 against 24.9. The 55.7 V/m that the withdrawn table attributed to "air within 2.5a" was the volume median, set by the end caps the figure does not show.

Why the conductor reads blue here. It is the colour range, not the physics. These figures keep the air in frame and must span the cut spike at 1.95 V/m, so the conductor — more than two orders of magnitude below it — lands at the bottom of the scale. A figure thresholded to the conductor alone and auto-ranged would show the same field spanning the whole colour map. Same data, different window.

The air's field is electrostatic, not **-j ω A** — that term is only 2.6e-04 V/m here. The loop carries 0.354 mV from one side of the cut to the other and that voltage appears across the surrounding air, strongest near the cut where the gradient is.

The 1.95 V/m at the cut is a mesh artefact, not a field. It equals the potential jump divided by the local element size to within 6 %, and the 30 mm mesh reads 1.72 V/m for the same ratio. It diverges under refinement. §5.8 does the arithmetic.

Every range in these figures is pinned, and on the 90 mm domain it has to be. An auto range is set by the single faintest cell in the frame, which sits in a far corner and moves whenever the domain or the grading changes — so the shipped figure and the control's could not honestly be laid beside each other. Pinning also keeps the cut spike from taking the whole colour map, which is how these figures first came out.

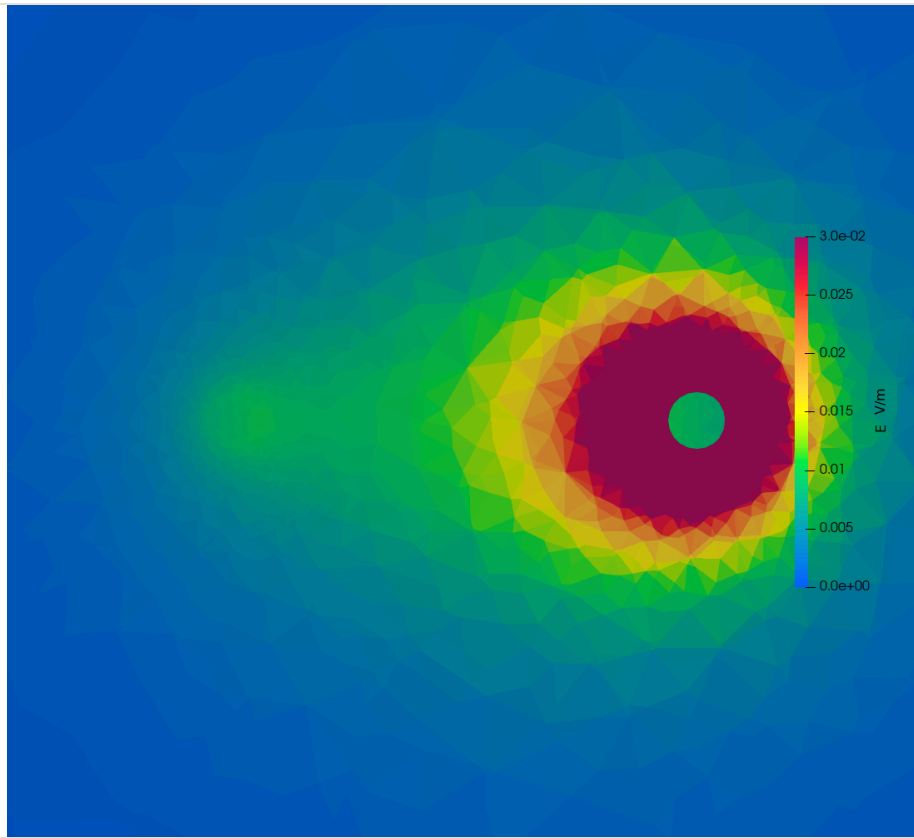
4.4 E on the plane normal to the torus, through the port

“Normal to the torus” at $\theta = 0$ means the plane whose normal is the centreline tangent there, which is $+y$ — so it is the $y = 0$ plane, and **that is the cut plane itself**. It slices the tube twice: at the port on the $+x$ side, and at $\theta = \pi$ on the $-x$ side.

How to read these. You are looking along $-y$ at a *vertical* plane containing the x and z axes. **$+x$ is to the right**. The plane crosses the tube twice — at $x = +6.5 \text{ mm}$, which is the **port**, and at $x = -6.5 \text{ mm}$, which is **just a joint** where the two half-rings meet with no potential difference across them. Only one of those two is a source, which is why only one is bright.

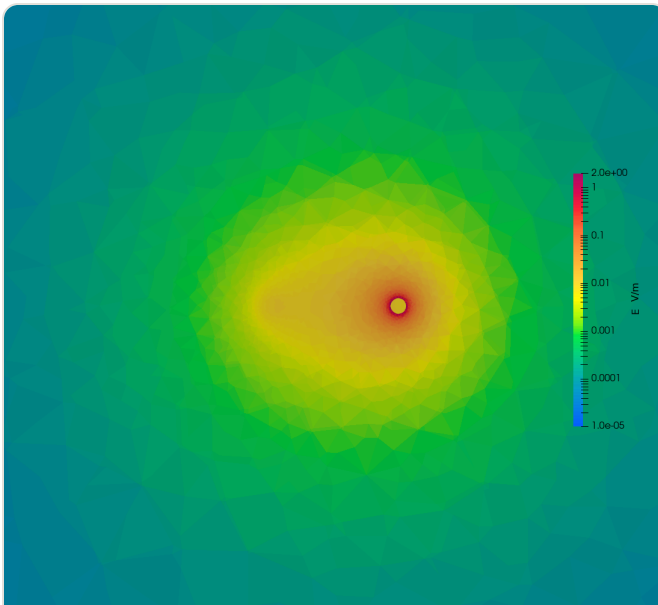
CROSSING	CONDUCTOR $ E $	AIR RIGHT AROUND IT
$x > 0$, $\theta = 0$ — the port	0.008669 V/m	median 0.1362 , max 1.9506 V/m
$x < 0$, $\theta = \pi$ — a plain joint	0.008641 V/m	median 0.0081, max 0.0109 V/m

Inside the two conductors $|E|$ is the **same** — 0.008669 against 0.008641 V/m, both J/σ , because the same current flows through both. Outside them it differs by **17x**. That difference *is* the port.

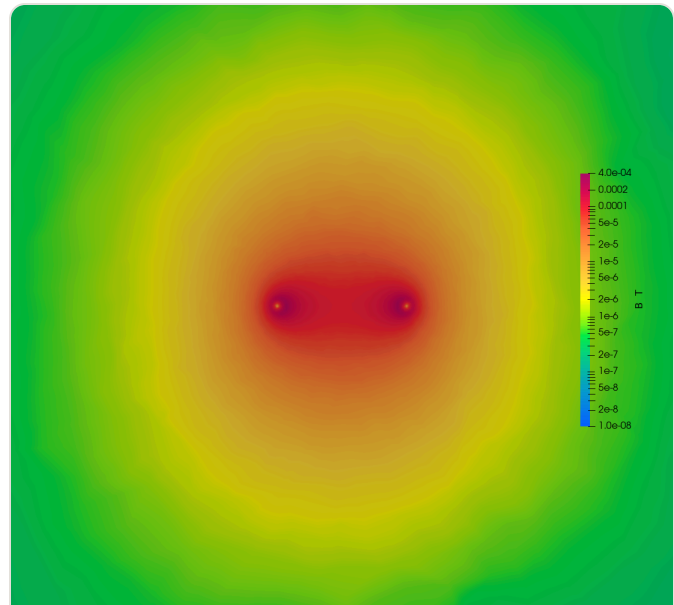


$|E|$ on the $y = 0$ plane, zoomed to the ring. The port is the bright crossing on the right; the faint patch on the left is the joint at $\theta = \pi$.

The same cut over the whole domain



|E| on the $y = 0$ cut, framed at 30 mm. Log scale over five decades: 1.95 V/m at the cut face down to $1.8\text{e-}05$ at the 90 mm wall. The cut itself is 180×90 mm now, and framing a square render on all of it shrinks the ring to a dot — so this view keeps the 30 mm window where the structure is, and the square **xy** views above are the ones framed on the whole domain.

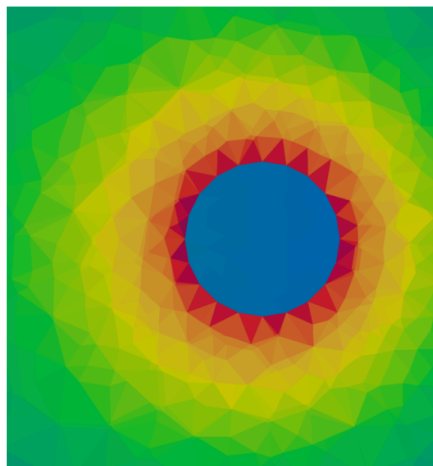


|B| on the same cut, for comparison. It is **continuous** across the port and shows both crossings equally — the asymmetry above is an **E** effect only.

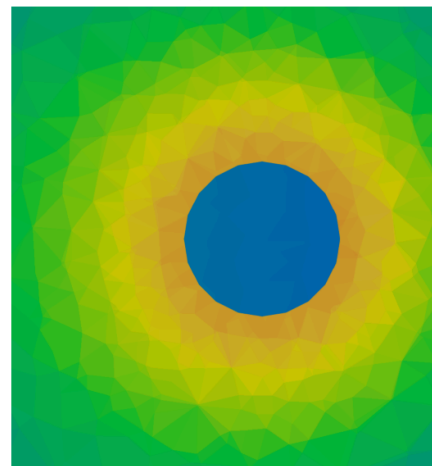
The **B** figure is the control. A magnetic field does not care that Φ jumps, so both tube crossings look alike there, and the two pictures side by side separate what the port does to **E** from what the geometry does to everything.

|E| at the port cross-section, same view and same log scale

ON the cut ($\theta = 0$)
air around the conductor reaches 1–2 V/m



one tube radius OFF the cut
the same air peaks near 0.2 V/m



Blue disc = the conductor, uniform at $J/\sigma = 8.6\text{e-}03$ V/m (no skin effect at $a/\delta = 0.086$). The difference between the two panels IS the port: Φ jumps 0.354 mV across a zero-thickness interface on the left, and is continuous on the right.

The measurement is the difference between these two. Same view, same tube, same log scale — the left panel lies *on* the cut, the right one tube radius off it. The red ring on the left is the port.

Each panel alone is an unremarkable blob, which is why they are drawn in one frame rather than merely adjacent: the eye has to compare them, and a shared colour bar is the only way that comparison means anything. The blue disc is the conductor, uniform at J/σ because $a/\delta = 0.086$ leaves no skin effect to resolve. `make_port_pair.py` composes it, sampling ParaView's own colour ramp out of the render rather than guessing a matplotlib equivalent.

This slice lies exactly on a discontinuity, which is the point. Φ jumps across the cut, so $E = -j\omega A - \nabla\Phi$ does too: the whole 0.354 mV appears across a zero-thickness interface, and the adjacent air sees an enormous gradient. B is continuous there and does not care, which is why §4.2's plot of the same plane needs none of this care.

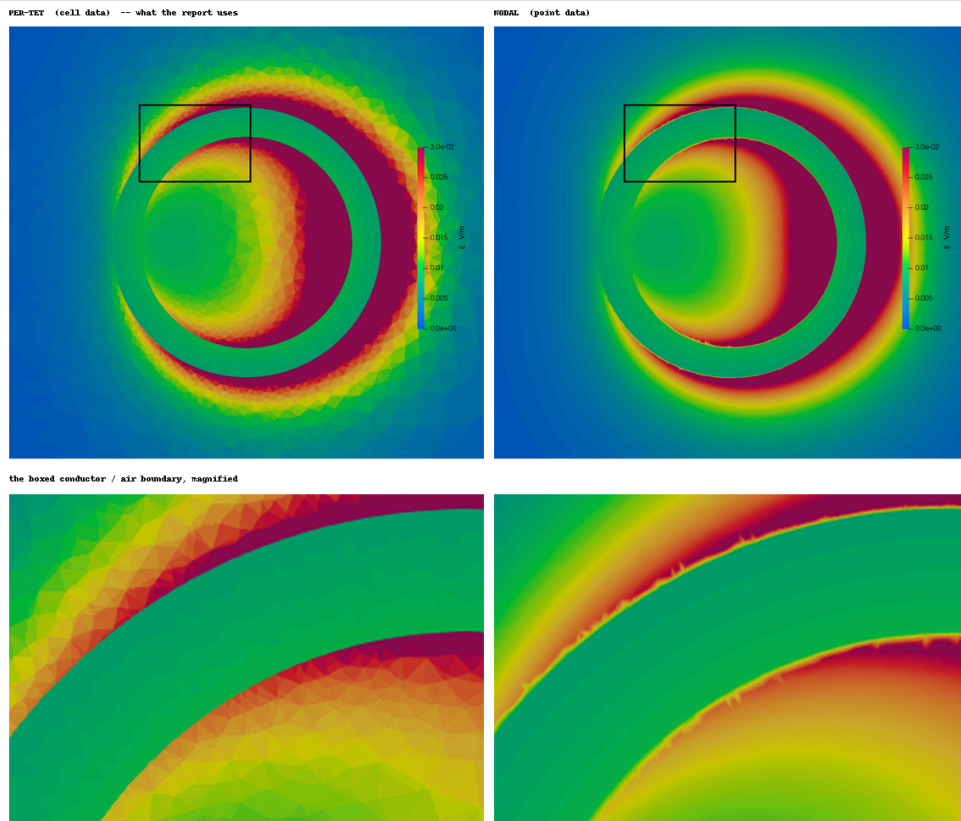
AIR SHELL HUGGING THE TUBE, $+x$ SIDE	MEDIAN $ E $
at the cut, $ y < 0.3a$	0.4041 V/m
one tube radius off, $0.7a < y < 1.3a$	0.1706 V/m
two to three radii round	0.0994 V/m
inside the conductor	0.008619 V/m (J/σ)
global maximum, at the cut face	1.9506 V/m — 226x the conductor

The field falls by 2.4x within one tube radius of the cut and by 4.1x by two or three. Both zooms are **log scaled**: the view spans more than two decades, from the conductor's $8.6e-03$ to 1.95 V/m, and a linear range that resolves the conductor saturates every air cell around it into one flat colour. The 226x is a property of the mesh, not of the ring — §5.8.

4.5 Nodal or per-tet, and why E is drawn per-tet

Every field here can be coloured from either of two arrays, and the choice is visible. **Nodal** (point) data is Gouraud-shaded: the colour ramps across each triangle between its vertex values, so the picture is smooth. **Per-tet** (cell) data is one flat fill per element, so every tetrahedron reads as a facet.

Smooth is better everywhere except at a material interface — and there it is actively wrong.



Same slice, same pinned range, same colour map. The only difference is which array the colour came from. Top row, the whole frame; bottom row, the boxed part of the conductor's inner wall at 3x.

What nodal averaging does. A nodal value is the volume-weighted average of the tetrahedra touching that node. For a node sitting *on* the conductor surface, those tets are on both sides of it: some hold the conductor's $8.7\text{e-}03 \text{ V/m}$, some the air's $4.2\text{e-}02$. The average is a blend of two numbers that are different **because the field genuinely jumps there** — E_t is continuous across the interface but E_n steps by the surface charge, so there is no single value at the wall to average towards.

Why it comes out ragged rather than merely blurred. Which tets happen to touch a given surface node varies around the ring, so the blend varies node to node. The result is the saw-tooth fringe in the bottom-right panel — the shape of the mesh, drawn as if it were the field. The per-tet panel beside it has the clean discontinuity the physics actually has, because every tetrahedron lies wholly in one material and nothing is averaged across anything.

The solver already knows which nodes these are. It writes a `material_interface` flag, and **14,736 nodes** — **5.0 % of the mesh** — **carry it**. That is exactly the set that produces the fringe.

BAND, DISTANCE FROM THE TUBE AXIS	NODAL <code> E </code>	PER-TET <code> E </code>	NODAL VS EXACT
0.80–0.90a, inside	8.503694e-03	8.521196e-03	–2.1 %
0.90–0.97a, inside	8.893547e-03	8.459322e-03	+2.4 %
1.03–1.10a, outside	4.251213e-02	4.109542e-02	–
1.10–1.30a, outside	3.776337e-02	3.786500e-02	–

against the exact interior value `I/(opK) = 8.684847e-03 V/m` at `p = R`. So the cost is a couple of per cent in the band next to the wall — not a disaster, and not the reason for the choice. **The reason is that a ragged fringe at the conductor surface is a picture of the mesh, in a report whose §5.8 is about what happens at exactly that surface.**

So: E is drawn per-tet, B is drawn nodal. `B` has no interface to fall over — `mu_r = 1` in both the ring and the air, so every tet contributes to every node and nothing is averaged across a jump. Its only cost is a far-field bias from the mesh grading, –7.5 % at the wall against –2.2 % per cell (`FIELD_POSTPROCESSING.md`), which matters for numbers and not for a picture.

None of this touches a validated number. Every figure in this report is illustration; every quantity in §5 is computed from the per-cell arrays by `analytic_comparison.py`, which averages nothing anywhere.

5. Comparison with the analytical solutions

Every number in this section comes from `analytic_comparison.py`, which prints the whole set in one run and derives each closed form in its docstring. Nothing here is hand-copied from a previous run.

```
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" analytic_comparison.py output
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" analytic_comparison.py output_small
```

What has a closed form here, and what does not. This matters more than it sounds, because the two places the fields look most dramatic — the cut, and the conductor's own surface — are the two places with the least to compare against.

QUANTITY	CLOSED FORM	STATUS
E inside the conductor	$E(\rho) = I/(\sigma\rho K)$	exact
R	$2\pi/(\sigma K)$	exact
B on the loop axis	superposition over the real current	exact
B anywhere in the air	filament, complete elliptic integrals	approximate, $O((a/R)^2) \approx 1.5\%$
L	$\mu_0 R [\ln(8R/a) - 2 + \frac{1}{4}]$	approximate, thin-wire
E in the air	—	none; only a bound at the wall, and §5.8 shows it is b
E at the cut	diverges	none exists; see §5.8

$K = \int dA/\rho$ is the one quantity every exact form is built on. It is taken off the mesh as $(1/2\pi) \int dV/\rho^2$, which assumes nothing whatever about the cross-section's shape — not that it is a circle, not that it is a 20-gon. For this mesh $K = 3.054193e-04 \text{ m}$.

5.1 The interior solution, derived

Current flows azimuthally, so try $\Phi = V\theta/2\pi$ and check it:

- **It conserves charge.** θ is harmonic — $\nabla^2\theta = 0$ in cylindrical coordinates — so $\nabla^2\Phi = 0$ and $\nabla \cdot \mathbf{J} = -\sigma\nabla^2\Phi = 0$.
- **No current leaks through the wall.** $\nabla\Phi = (V/2\pi)(1/\rho)\hat{\phi}$ is purely azimuthal, and the tube wall is a surface of revolution whose normal lies in the $\rho - z$ plane. So $\partial\Phi/\partial n = 0$ on it identically.

Both conditions hold exactly, for any cross-section shape, so this **is** the solution — not an approximation to it. Therefore

$$E(\rho) = V/(2\pi\rho) \qquad J(\rho) = \sigma V/(2\pi\rho)$$

E is not uniform across the section; it goes as $1/\rho$. Every filament at cylindrical radius ρ closes a path of length $2\pi\rho$, and all filaments share the same terminal voltage V , so the field along each is V divided by its own path length. The inner filaments have a shorter path, so they carry more current: the current *crowds toward the inner radius*. Across this tube the ratio is exactly

$$E(R-a)/E(R+a) = (R+a)/(R-a) = 7.3/5.7 = 1.2807$$

Integrating J over the section gives $I = \sigma VK/2\pi$, hence $V = 2\pi I/(\sigma K)$ and the two forms used below:

$$E(\rho) = I/(\sigma\rho K) \qquad R = V/I = 2\pi/(\sigma K)$$

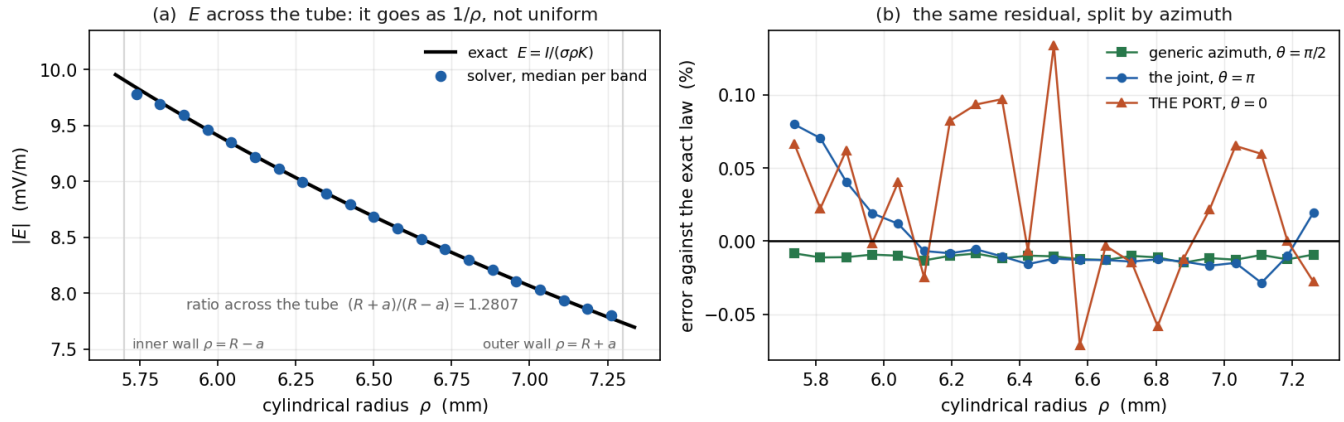
This holds while there is no skin effect to redistribute the current. Here $a/\delta = 0.086$, so there is none — case 04 is the case where that assumption is deliberately broken.

5.2 **E** inside the conductor — exactly where it was compared

The comparison is made cell by cell, in the conductor only. Specifically: every tetrahedron with `body_tag = 1`, taking its complex `E_cell_real / E_cell_imag` at the tet centroid, forming $|E| = \sqrt{Re^2 + Im^2}$, and dividing by $I/(\sigma\rho K)$ evaluated at that same centroid's cylindrical radius ρ . The cells are then binned by ρ and the median and 90th percentile of the relative error are reported per band. **No comparison is made in the air** — there is no closed form there (§5.8).

The reason for splitting by azimuth as well as radius is that the discretisation has to carry a *discontinuity* in Φ at the cut, and the question is whether the solution is worse there.

Case 05: E inside the conductor against the exact interior solution ($K = \int dA/\rho$ taken from the mesh)



Panel (a) is the $1/\rho$ law with the solver's per-band medians on top of it. **Panel (b)** is the same data as a residual, split three ways: at the port, at the joint diametrically opposite it, and at a generic azimuth that is neither.

REGION	P BAND	CELLS	MEDIAN ERROR	90TH PCT ERROR
whole conductor	0.88–0.94R	4500	−0.013 %	0.035 %
	0.94–1.00R	6016	−0.015 %	0.030 %
	1.00–1.06R	6268	−0.016 %	0.027 %
	1.06–1.12R	6625	−0.016 %	0.029 %
at the port, $\theta = 0$	0.88–0.94R	188	+0.033 %	0.801 %
	0.94–1.00R	187	+0.062 %	0.392 %
	1.00–1.06R	187	+0.027 %	0.965 %
	1.06–1.12R	165	+0.013 %	2.404 %
at the joint, $\theta = \pi$	0.88–0.94R	182	+0.025 %	0.153 %
	0.94–1.00R	182	−0.010 %	0.020 %
	1.00–1.06R	178	−0.013 %	0.021 %
	1.06–1.12R	183	−0.015 %	0.053 %
generic azimuth, $\theta = \pi/2$	0.88–0.94R	343	−0.010 %	0.021 %
	0.94–1.00R	460	−0.011 %	0.021 %
	1.00–1.06R	469	−0.012 %	0.020 %
	1.06–1.12R	503	−0.011 %	0.022 %

Over the whole conductor the field is within 0.016 % of the exact law, and the 90th percentile is 0.035 % — that is the solver reproducing a closed form to four significant figures, everywhere, in all 23,409 conductor elements. The four bands partition the conductor exactly; nothing is left out of them.

The port costs about a factor of a hundred, and only in the spread. Its *medians* are still at the 0.01–0.06 % level, so there is no bias; what grows is the scatter, from a 90th percentile of 0.02 % at a generic azimuth to 2.40 % next to the cut. Narrowing to the cells that actually touch the cut face:

CELLS TOUCHING THE CUT FACE ($ Y < 0.12 \text{ MM}$, $X > 0$)	
count	353
median error	+0.058 %
worst single cell	−3.378 %

That is the honest cost of representing a discontinuity with a finite element: one layer of cells gets it wrong by a few per cent, the median is unbiased, and one element further in it is gone. The joint at $\theta = \pi$ — geometrically identical, two half-rings meeting, but with no potential difference across it — sits at 0.02 %, which confirms the error belongs to the *cut*, not to the mesh seam.

5.3 R − 0.13 %

measured	3.542415e-04 Ω
exact for this shape, $2\pi/(\sigma K)$	3.546953e-04 Ω
error	−0.128 %

The exact value is not $2\pi R/(\sigma A)$. That formula assumes every filament has the same path length, and §5.1 shows they do not. Using it gives 3.560434e-04 Ω , +0.380 % — three times the solver's own error, and in the opposite direction. The naive form is wrong because it ignores the $1/\rho$ crowding.

The 2.5 % gap to the reference tool decomposes exactly: +1.61 % because our cross-section is a 20-gon with 1.64 % less area than a circle, and 0.91 % between the reference and the exact round-torus value.

5.4 B on the loop axis — exact, no filament approximation

On the axis there is a closed form that uses the **real** current distribution, not a thin-wire idealisation. A circular filament of radius ρ carrying I_f produces, on its own axis at axial distance d ,

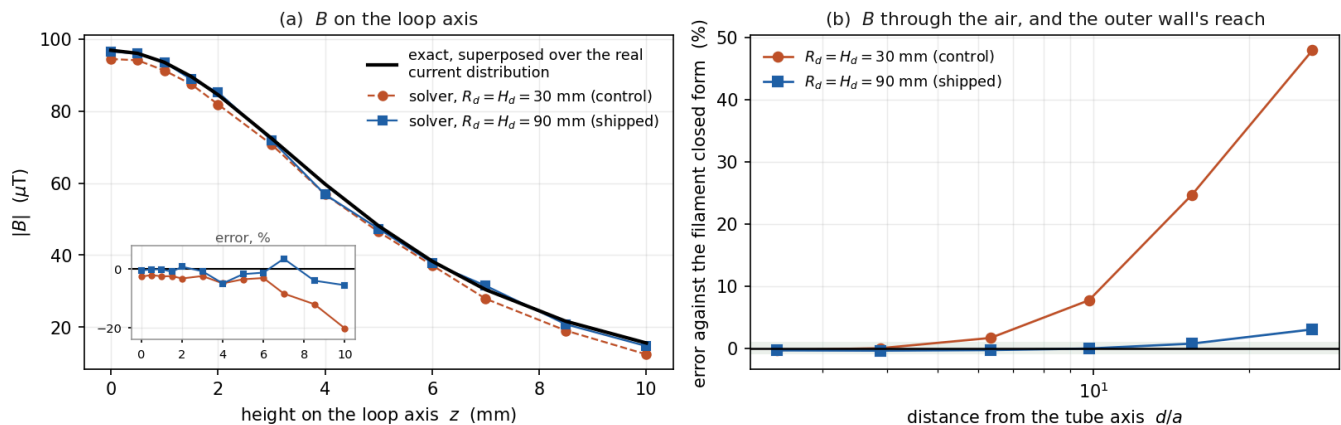
$$B = \mu_0 I_f \rho^2 / (2 (\rho^2 + d^2)^{3/2})$$

Superposing the tube's filaments, each carrying $J(\rho) dA = I dA/(\rho K)$, and writing $dA = dV/2\pi\rho$:

$$B_z(z_0) = (\mu_0 I / 4\pi K) \int dV / (\rho^2 + (z_0 - z)^2)^{3/2}$$

The integral runs over the conductor cells of the mesh, so this reference knows the actual cross-section and the actual $1/\rho$ current profile. It is exact.

Case 05: B against its two closed forms — the far-field error is the OUTER BOUNDARY, not the mesh



Z_0 (MM)	MEASURED (T)	EXACT (T)	ERROR	ERROR, 30 MM CONTROL	FILAMENT $M_0 I$
0.0	9.648862e-05	9.684792e-05	-0.371 %	-2.512 %	9.666439e-05
1.0	9.347581e-05	9.349618e-05	-0.022 %	-2.426 %	9.333132e-05
2.0	8.522485e-05	8.451782e-05	+0.837 %	-3.155 %	8.439933e-05
3.0	7.191648e-05	7.241929e-05	-0.694 %	-2.329 %	7.235454e-05
5.0	4.732007e-05	4.812508e-05	-1.673 %	-3.434 %	4.813579e-05
7.0	3.150876e-05	3.041954e-05	+3.581 %	-8.317 %	3.045491e-05
10.0	1.476315e-05	1.561401e-05	-5.449 %	-20.191 %	1.564690e-05

Two things to read off this. First, the shipped column **scatters either side of zero** while the 30 mm control is **one-signed and grows monotonically** — that is the difference between discretisation noise and a systematic boundary error, and it is the heart of §5.7. Second, the last column shows the **filament idealisation is worth 0.19 %** at the centre (9.666439e-05 against the exact 9.684792e-05): at $a/R = 0.123$ the tube's thickness barely matters, which is what licenses §5.5.

5.5 B through the air — the filament's elliptic-integral form

Away from the wire the ring can be treated as a filament of radius R , and then B is known in closed form **everywhere**, not just on the axis, through the complete elliptic integrals $K(m)$ and $E(m)$:

$$Q = (R+p)^2 + z^2 \quad m = 4Rp/Q \quad D = (R-p)^2 + z^2$$

$$B_z = (\mu_0 I / 2\pi) Q^{-1/2} [K(m) + E(m) (R^2 - p^2 - z^2) / D]$$

$$B_\rho = (\mu_0 I / 2\pi) (z/p) Q^{-1/2} [-K(m) + E(m) (R^2 + p^2 + z^2) / D]$$

K and E are evaluated by the arithmetic–geometric mean, which reaches machine precision in about five iterations. **The script checks this against itself before using it:** as $p \rightarrow 0$ the pair must collapse to the on-axis formula $\mu_0 I R^2 / 2 (R^2 + z^2)^{3/2}$, and it does, to $1.8e-14$ relative.

Unlike §5.1–5.4 this form is an **approximation**. It ignores the tube's thickness, so it errs $0((a/R)^2) \approx 1.5 \%$, and it *diverges* at the ring. It is therefore used only outside $2a$ from the tube axis — and §5.4 measured how good it is there: 0.19 %.

The comparison is again cell by cell: every air cell (`body_tag = 2`) outside `2a` , binned by distance `d` from the tube axis.

D/A	CELLS	MEDIAN B (T)	FILAMENT (T)	MEDIAN ERROR	30 MM CONTROL	MEDIAN h/d
2–3	48237	8.940967e-05	8.990699e-05	–0.36 %	–0.31 %	0.21
3–5	33704	4.936850e-05	4.960650e-05	–0.40 %	–0.01 %	0.22
5–8	16614	2.121133e-05	2.127459e-05	–0.31 %	+1.64 %	0.23
8–12	8774	8.940558e-06	8.931514e-06	–0.06 %	+7.70 %	0.23
12–20	7567	3.547741e-06	3.540672e-06	+0.73 %	+24.69 %	0.24
20–35	6194	1.029378e-06	1.018296e-06	+3.02 %	+47.92 %	0.24

From `2a` out to `20a` the solver tracks the closed form inside $\pm 0.75\%$, across four decades of field magnitude — $8.9\text{e-}05$ down to $3.5\text{e-}06$ T. The $+3.02\%$ in the last band is the 90 mm wall, three times further out than the control's, doing exactly what the control's did at 20–35a.

The `h/d` column is in the table to rule out element size as the cause: it is flat at $\approx 0.21\text{--}0.24$ across every band, so resolution relative to the distance being resolved is constant while the control's error grows by a factor of 150. That is not discretisation.

5.6 **L** — the closed form, and two retractions

For a thin circular loop in free space,

$$L = \mu_0 R \left[\ln(8R/a) - 2 + \frac{1}{4} \right]$$

The bracket is the external inductance from the loop's own flux ($\ln(8R/a) - 2$) plus the internal term $\frac{1}{4}$ for uniform current inside the wire. With `R = 6.5 mm` and `a = 0.8 mm` , $\ln(8R/a) = \ln 65$ and `L = 19.803 nH` . Using the *equal-area* radius of the 20-gon instead, `a_eq = 0.79343 mm` , gives 19.870 nH — so the choice of which radius to call `a` is itself worth 0.34 %, and no comparison here is meaningful below that.

	L	VS CLOSED FORM (A = CIRCUMRADIUS)
measured, shipped (90 mm)	19.785 nH	-0.09 %
measured, 30 mm control	19.377 nH	-2.15 %
reference tool	19.748 nH	-0.27 %
closed form, free space	19.803 nH	—

Two statements previously made here were wrong, and both are retracted.

The first: *"refining the tube moved L the wrong way, so resolution is not the cause."* That test refined `lc_ring` globally while half the ring was still being meshed coarse by the $+\pi / -\pi$ sweep bug of §6.2, so it was measuring the bug.

The second: *"the domain is ruled out — doubling it to 60 mm moved L by +0.2 %."* That mesh still carried the same sweep bug, **and** it coarsened the far field so much that elements near the loop grew from 4 mm to 8.8 mm. Two variables, neither controlled.

Both are negative results produced by a test that moved more than one thing at a time, and both pointed away from the actual cause for months. That is the lesson worth keeping from this case, rather than anything about loops.

Extracting L here is ill-conditioned. $\omega L/R = 0.017$: the reactance is 1.7 % of the resistance, so $\text{Im}(Z)$ is a small difference between large numbers. Dividing by the *measured* current instead of the imposed 1 A lets that current's own 0.02 % imaginary part into the answer and moves L by 1.3 % — fifty times its own size. `verify_loop.py` divides by the imposed current.

5.7 Why the domain is 90 mm

`[boundary] outer = flux_tangential` sets $n \cdot B = 0$ on the domain wall. That confines the return flux inside the domain and behaves like a **flux-excluding shell** — equivalently, an image loop carrying opposing current. Three consequences follow from that one picture, and all three are observed, with the signs it predicts:

- flux is pushed **out** of the middle of the ring → B on the axis reads **low**;
- less flux links the loop → L reads **low**;

- the excluded flux has to go somewhere, and it crowds **along** the wall → **B** out near the wall reads **high**.

loop_small.geo is the 30 mm domain this case used to ship, kept as the control that demonstrates it. The grading is scaled with the domain so that element size **on the loop axis matches to within 10 %** out to **z = 10 mm**: the only thing that differs between the two meshes is where the wall is.

OBSERVABLE	30 MM CONTROL	90 MM, SHIPPED
B on the axis, at the loop centre	-2.51 %	-0.37 %
L	-2.15 %	-0.09 %
B in the air, d = 8-12a	+7.70 %	-0.06 %
B in the air, d = 12-20a	+24.69 %	+0.73 %
B in the air, d = 20-35a	+47.92 %	+3.02 % ← <i>the new wall</i>

At **R_d = 30 mm** the wall is 2.3 loop radii out, which is simply too close. Panel (b) of §5.4's figure is the whole argument in one frame.

And the larger domain is cheaper. **Threshold** grades linearly in distance, so raising **lc_far** and **d_far** with the domain reproduces the old grading wherever the fields actually are, while putting far fewer elements in the empty mid-field:

	90 MM, SHIPPED	30 MM CONTROL
tets	217,719	223,879
unknowns	509,598	524,050
MUMPS fill-in	674.1 M	724.0 M

27x the volume for 3 % fewer elements and 7 % less fill.

Wall-clock is a wash and should not be quoted either way. Back to back on an idle machine, 43.5 s against 43.8 s. An earlier measurement here said 103 s against 130 s; that pair was taken with gms and ParaView running alongside, and the 1.3x it appeared to show was machine load, not the mesh. The three counts above are properties of the mesh. The seconds are a property of whatever else was running.

5.8 Where there is no closed form

Two places in this problem have no analytic value to compare against, and both are places the pictures in §4 draw the eye to. One of them also turned out to be barely measurable, which is a correction to an earlier version of this section.

At the tube wall there is a bound, and this mesh can only just see it

$E_{\text{tangential}}$ is continuous across any interface and E_{normal} jumps by the surface charge, so $|E|$ just outside a current-carrying conductor is $\geq |E|$ just inside — always, pointwise.

An earlier version of this report tested that with a shell statistic and declared it confirmed. The statistic does not test the bound. It compared the median over 0.95a–0.99a inside against the median over 1.01a–1.05a outside — different points, from samples of a few dozen cells. On the 30 mm mesh it reads 1.086; on the shipped 90 mm mesh, same physics, it reads 0.984. A statistic that changes sides when the far-field boundary moves was never measuring a local boundary condition.

The air's own field is not even monotonic across those shells, which is why a thin band is a bad sample:

SHELL OUTSIDE THE WALL, $\theta = \pi \pm 14^\circ$	CELLS	MEDIAN $ E $
1.01–1.05a	99	8.338984e-03
1.05–1.10a	497	9.325833e-03
1.10–1.20a	620	8.865476e-03
1.20–1.40a	806	8.377270e-03

The bound is pointwise, so it has to be tested pairwise. Taking each near-wall conductor cell and its *nearest* air cell:

PAIRED ACROSS THE WALL, $\theta = \pi \pm 14^\circ$

pairs	577
median centre separation	0.153a
$ E_{\text{air}} / E_{\text{cond}} $, median	1.026
10th / 90th percentile	0.971 / 1.166
fraction ≥ 1	67 %

That is as much as this discretisation can say, and the report should not claim more. **E**

is piecewise constant per tet; the element size at the wall is 0.343a; the paired centres straddle the interface at about $\pm 0.08a$; and the air's field varies by $\sim 10\%$ over that distance — comparable to the jump being looked for. The median is above 1 and two thirds of pairs are, which is consistent with the bound and too coarse to be a proof of it. Individual pairs below 1 are that averaging, not a violation.

Note also that "the field is highest inside the conductor" is **not** a law. The opposite is, and even that is only a bound: its *value* would require solving the exterior problem, which is what the FEM is for and which there is nothing to check against.

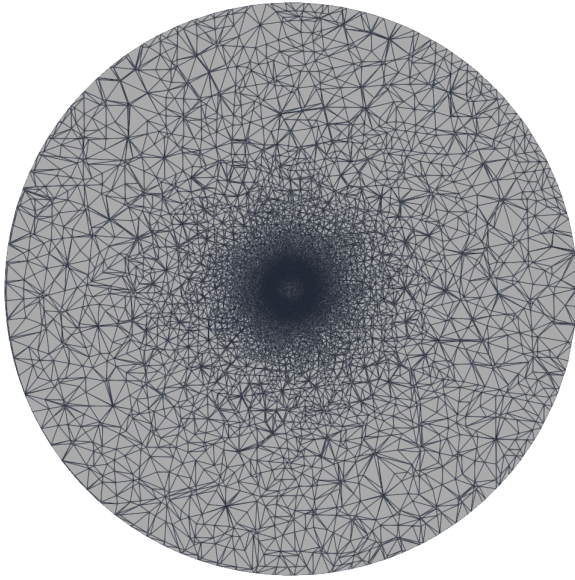
At the cut the analytic answer is that **E diverges**

A finite voltage across a zero-thickness interface is a singularity: there is no finite number there, and what the solver returns is set by the local element size.

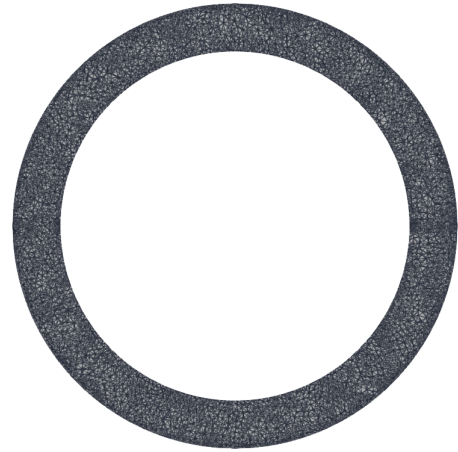
	SHIPPED (90 MM)	30 MM CONTROL
peak $ E $ in air	1.9506 V/m	1.7218 V/m
local element size h	0.1928 mm	0.2001 mm
V/h	1.8397 V/m	1.7724 V/m
ratio peak / (V/h)	1.06	0.97

Two different meshes, two different peak values, the same ratio to V/h . The spike *is* the potential jump spread over one element. It scales as $1/h$ and never converges, so quoting "1.72 V/m, 200x the conductor" as a result — as an earlier version of this report did — is meaningless without naming the mesh. Refine the cut region and the number goes up.

6. The mesh, and two bugs found building it



Whole domain, graded from 0.25 mm at the ring to 4 mm at the wall.



The ring, about 5 elements across the 1.6 mm tube.

6.1 An OCC Torus will not mesh with a cut in it

The obvious construction — an **OpenCASCADE Torus**, cut, fragmented against the air — fails with *“Invalid boundary mesh (overlapping facets)”*: a curved torus face overlapping itself. That was tried **eight ways**: partial tori welded with **Coherence**, cuts by oversized discs, by exact-size discs, by thin boxes, cuts moved off the parametric seam, the seam rotated away from the cuts, the air fragmented before and after the cut. Every one failed identically.

Revolving a polygon has none of that trouble, and it is what this project does for every other conductor anyway, because planar lateral faces make $\partial\Phi/\partial n = 0$ hold exactly.

Note also that a ring needs **two** cuts. Cut in one place it is still one connected volume, because the current can go round the other way.

6.2 One $+\pi$ sweep and one $-\pi$ sweep is wrong in a way that is easy to miss

The ring was first revolved as one **+Pi** extrude and one **-Pi** extrude. It meshes, it passes every check, and **both halves come out with the right volume** — 40.386 against 40.307 mm³, equal to 0.2 %. But:

	TETS	MEAN H	VOLUME
+Pi half	11,621	0.302 mm	40.386 mm ³
-Pi half	4,557	0.373 mm	40.307 mm ³

Half the conductor was quietly meshed two and a half times coarser. It showed up as a ragged lower half in the field plots, and was measured before it was believed. Chaining two *positive* sweeps — the second starting from the first's end face — gives a count ratio of 0.988.

And it moved the physics. On the 30 mm domain of the day, **L** went 19.038 → 19.377 nH, closing 1.87 of the 4.02 points of the gap to theory. It also **retracts** an earlier conclusion recorded in this case: that refining the tube moved **L** the wrong way, so resolution was not the cause. That test refined **lc_ring** globally while half the ring stayed coarse, so it was measuring the wrong thing. Resolution *is* part of it.

7.1 The cut's local field is tree-dependent — the assertions are not

Everything this case asserts is an **integral**, and that turns out to be the only reason it passes.

Running **07_GaugeInvariance**'s permutation harness on this case — the same problem solved on three different spanning trees, which changes nothing physical — gives:

	BASE → PERMA	BASE → PERMB
L , asserted 19.7854 ± 0.05 nH	+0.0035 nH	+0.0009 nH
 V 	~3e-6 relative	~3e-6 relative
J within 2 mm of the cut	8.3 %	—
B within 2 mm of the cut	7.4 %	—

and the change dies away from the cut: **J** moves 8.3 % inside 2 mm, **2.3e-4** at 2–5 mm and **7.2e-6** at 5–10 mm.

It is not numerical noise. All three solves are exact (backward errors 9.1e-23, 9.0e-23, 9.2e-23); $\Delta E = \Delta J / \sigma$ to a ratio of 1.0000, so the **E** and **J** figures are one physical change reported against different global maxima; and it decays smoothly over some 11,000 nodes

rather than sitting on the 205 cut nodes, so it is not an artifact of which side of the cut the output happens to print.

So this case is correct and its tolerances are honest. `L` moves by 0.018 % of itself, which is 7 % of a tolerance it was never close to failing.

What is not safe is the local field at the feed. The current distribution within a couple of millimetres of the cut depends on which spanning tree the gauge happened to pick. Nothing in `expected.txt` could catch that, because every assertion here is an integral and integrals average the discrepancy away.

Read the right thing off this case. `R`, `L`, `Z`, the current through a plane, and the fields away from the cut are all sound. `J` and `B` within a few millimetres of the cut are not — treat them as indicative, not quantitative, until the gauge is replaced.

This is the measurement that sets the bar for that replacement: a new gauge has to make the local field near an interior port tree-independent, not merely keep `L` inside tolerance. See `docs/GAUGE_CHOICE.md` §1.3, where it is recorded alongside the literature and the practitioner experience it refines.

8. What is asserted automatically

`check.bat` runs this case through `verify_loop.py`, not `../verify.py`: the rod verifier finds the drive by differencing Φ between two end caps, and a ring has none. A case-local `verify_*.py` wins where one exists.

`|I|`

through
the cut 1.000503 vs 1.0, tol 1e-3

`|I|` at

$\theta = \pi/2$ 1.000176 vs 1.0, tol 1e-3

`R` vs

exact
for this
shape 0.99872 vs 1.0, tol 2e-3

`L`

19.7854 nH, tol 0.05 — now **0.09 % from the closed form**, so for the first time this line is a vali

Reproducing

```
gmsht loop.geo -3 -o loop.msh
..\run_case.bat loop_50hz.aphi
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" make_plots.py output
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" make_port_pair.py
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" verify_loop.py
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" analytic_comparison.py output
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" make_analytic_plots.py
sh build_doc.sh
```

And the 30 mm control, which §5.7 rests on. `make_analytic_plots.py` has to be re-run after it for the second curve in `fig/analytic_B.png` to appear:

```
gmsht loop_small.geo -3 -o loop_small.msh
..\run_case.bat loop_50hz_small.aphi
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" analytic_comparison.py output_small
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" make_analytic_plots.py
```

`loop_small.geo` is the 30 mm domain this case shipped until the boundary turned out to be the largest error in it. It is kept because §5.7's comparison is the evidence for the 90 mm, and a control nobody can rerun is not evidence. **Run `build_doc.sh` after regenerating figures**, or this document keeps showing the previous run's pictures — they are baked in.